

Problem 8.10. The Japanese game *go-moku* is played by two players, “X” and “O” on a 19×19 grid. Players take turns placing markers, and the first player to achieve five of her markers consecutively in a row, column, or diagonal is the winner. Consider this game generalized to an $n \times n$ board. Let

$GM = \{\langle B \rangle \mid B \text{ is a position in generalized go-moku, where player “X” has a winning strategy}\}.$

By a *position* we mean a board with markers placed on it, such as may occur in the middle of a play of the game, together with an indication of which player moves next. Show that $GM \in PSPACE$.

Proof.

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